Dr. Sergey Yurkevich, MSc

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Curriculum Vitae

Personal Information

Date of Birth	July 10, 1996 (29 years old)
Place of Birth	Moscow, Russian Federation
Nationality	Austrian

Research experience

01/10/2023–present **Researcher for safe AI and computer vision**,
In: Team **CSAI** (Center for Safe Algorithmic Intelligence) *AI Alligators*,
At: A&R TECH, Austria

01/08/2023–31/12/2023 **Postdoctoral researcher in mathematics**,
In: Research project *Algebraic Solutions of Fuchsian Differential Equations*, FWF P-34765,
At: University of Vienna, Austria

01/09/2021–31/07/2023 **Predoctoral researcher and PhD student**,
In: Doctoral research project *Integer sequences, algebraic series and differential equations*,
At: University of Vienna, Austria

01/02/2020–31/08/2021 **Predoctoral researcher and PhD student**,
In: Research project *Algebraic Solutions of Fuchsian Differential Equations*, FWF P-34765,
At: University of Vienna, Austria

01/09/2020–06/07/2023 **Predoctoral researcher and PhD student**,
At: *SpecFun/MathExp*, Inria Saclay, France

01/09/2020–06/07/2023 **Predoctoral researcher**,
At: *De rerum natura*, ANR reserach project ANR-19-CE40-0018, France

01/09/2018–31/01/2020 **Student employee**,
In: research project *Artin Approx., Arc Spaces & Resolution of Singularities*, FWF P-31338,
At: University of Vienna, Austria

Education

01/02/2020–06/07/2023 **PhD in Mathematics**,
Advisors: Alin Bostan and Herwig Hauser,
Thesis: Integer sequences, algebraic series and differential operators.
University of Vienna, Faculty of Mathematics, Austria
Université Paris-Saclay and Inria Saclay, France

01/10/2018–15/09/2020 **Master of Science in Quantitative Finance**,
Advisor: Kurt Hornik.
Vienna University of Economics and Business (WU Vienna),
Faculty of Quantitative Finance, Austria

01/10/2017–31/01/2020 **Master of Science in Mathematics**,
Advisor: Herwig Hauser.
University of Vienna, Faculty of Mathematics, Austria

01/10/2015–24/05/2018 **Bachelor of Science in Business Administration**.
Vienna University of Economics and Business (WU Vienna), Faculty of Economics and Social Sciences, Austria

01/10/2014–07/09/2017 **Bachelor of Science in Mathematics**.
University of Vienna, Faculty of Mathematics, Austria

01/09/2006–15/06/2014 **High School**.
Bundesrealgymnasium GRG3 Radetzkystraße, Vienna, Austria

01/09/2006–15/06/2014 **High School**.
Russian Embassy School, Vienna, Austria

Scholarships

01/01/2024–31/12/2025 **WTZ Austria-France Collaboration FR 02/2024**,
Title: "Differential Equations and Galois Groups",
Total funding: 9.000€,
Agency for Education and Internationalisation, Austria.

01/09/2021–31/07/2023 **ÖAW DOC Fellowship P-26101**,
Title: "Integer sequences, algebraic series and differential equations",
Total funding: 80.000€,
Austrian Academy of Sciences, Austria.

29/08/2023–27/09/2023 **Bourse d'Excellence (Excellence scholarship)**,
Mobility scholarship: Austria/France.
Total funding: 1.704€,
Institut français d'Autriche, Austria.

01/01/2021–31/12/2022 **WTZ Austria-France Collaboration 46411Y & FR09/2021**,
Total funding: 8.000€,
Agency for Education and Internationalisation, Austria.

01/02/2018–31/01/2024 **Scholarship for young talented researchers**,
Total funding: 3.000€,
Pro Scientia, Austria.

01/10/2015–01/10/2020 **Scholarship of excellence (Leistungsstipendium)**,
Total funding: 5.000€,
University of Vienna, Austria.

01/10/2016–01/10/2020 **Scholarship of excellence (Leistungsstipendium)**,
Total funding: 5.000€,
Vienna University of Economics and Business, Austria.

Competitions and Honors

4WARD **Third Uniqa 4WARD Math Challange**.
"Third Place", 500€ — Vienna, 01/06/2020.

NCUMC **North Countries Universities Mathematical Competition**.
"First Prize" — St. Petersburg, 15/04/2018.

IMC **International Mathematical Competition**.
"Honourable Mention" — Bulgaria, 06/08/2017,
"Third Prize" — Bulgaria, 02/08/2015, 31/07/2016 & 28/07/2018.

VJIMC **Vojtěch Jarník International Mathematical Competition**.
"Honourable Mention" — Czech Republic, 27/03/2015, 08/04/2016 & 29/03/2019,
"Third Prize" — Czech Republic, 01/04/2017 & 13/04/2018.

IMO **International Mathematical Olympiad**.
"Third Prize" (Bronze medal) — South Africa, 09/07/2014.

MEMO **Middle European Mathematical Olympiad**.
"Honourable Mention" — Hungary, 22/08/2013.

Publications

Published and accepted articles

2025 "**Systems of Discrete Differential Equations, Constructive Algebraicity of the Solutions**", with H. Notarantonio.
In: *Combinatorial Theory*, vol 5(2), pp. 1–34.
DOI: doi.org/10.5070/C65265403

2024 "**Algebraicity of hypergeometric functions with arbitrary parameters**", with F. Fürnsinn.
In: *Bulletin of the London Mathematical Society*, vol 56, pp. 2824–2846.
DOI: doi.org/10.1112/blms.13103

December 2023 "**On the representability of sequences as constant terms**", with A. Bostan and A. Straub.
In: *Journal of Number Theory*, vol 253, pp. 235–256.
DOI: doi.org/10.1016/j.jnt.2023.06.015

July 2023 "**Beating binary powering for polynomial matrices**", with A. Bostan and V. Neiger.
In: Proceedings of ISSAC'23, pp. 70–79, ACM Press.
DOI: doi.org/10.1145/3597066.3597118

July 2023 "**Algebraicity for solutions of systems of functional equations**", with H. Notarantonio.
In: Séminaire Lotharingien de Combinatoire , vol. 89B-13, pp. 1–12, 2023. Talk at FPSAC'23.
DOI: https://www.mat.univie.ac.at/~slc/

April 2023 "**On the q-analogue of Pólya's Theorem**", with A. Bostan.
In: *Electronic Journal of Combinatorics*, vol 30, pp. 2.9:1-9.
DOI: doi.org/10.37236/11269

March 2023 "**Fast Computation of the N-th Term of a q-Holonomic Sequence and Applications**", with A. Bostan. In *Journal of Symbolic Computation*, vol 115, pp. 96–123.
DOI: doi.org/10.1016/j.jsc.2022.07.008

January 2023 "**An algorithmic approach to Rupert's problem**", with J. Steininger,
In: *Mathematics of Computation*, vol 92, pp. 1905–1929.
DOI: doi.org/10.1090/mcom/3831

June 2022 "**Extended abstract for: solving Rupert's problem algorithmically**", with J. Steininger.
In: *ACM Communications in Computer Algebra*, vol 56, issue 2, pp. 32–35.
DOI: doi.org/10.1145/3572867.3572870

September 2022 "**The art of algorithmic guessing in gfun**".
In: *Maple Transactions*, vol 2, No. 1, pp. 14421:1–14421:19.
DOI: doi.org/10.5206/mt.v2i1.14421

February 2022 "**A hypergeometric proof that** Iso **is bijective**", with A. Bostan.
In: *Proceedings of the American Mathematical Society*, vol 150, pp. 2131–2136.
DOI: doi.org/10.1090/proc/15836

February 2022 "**On a Class of Hypergeometric Diagonals**", with A. Bostan.
In: *Proceedings of the American Mathematical Society*, vol 150, pp. 1071–1897.
DOI: doi.org/10.1090/proc/15693

May 2021 "**Diagonal Representation of Algebraic Power Series: A Glimpse Behind the Scenes**".
In: Transcendence in Algebra, Combinatorics, Geometry and Number Theory (TRANS 2019). *Springer Proceedings in Mathematics and Statistics*, vol 373, pp. 309–339.
DOI: doi.org/10.1007/978-3-030-84304-5_13

Preprints

August 2025 "**A convex polyhedron without Rupert's property**", with J. Steininger.
arXiv: arxiv.org/abs/2508.18475.

November 2024 "**Isoperimetric Ratios of Toroidal Dupin Cyclides**", with A. Bostan and T. Yu.
arXiv: arxiv.org/abs/2411.05643.

September 2022 "**On the First Fundamental Theorem for $\mathrm{GL}_2(K)$ and $\mathrm{SL}_2(K)$**", with H. Melánová.
arXiv: arxiv.org/abs/2009.08904, 9 pages.

Research talks and participation at conferences

For **slides** and **recordings** see: *yurkevi.ch/me.html#presentations*.

Invited talks at conferences

27/07/2023 "**Beating binary powering for polynomial matrices**",
at Internat. Symposium on Symbolic & Algebraic Computation *(ISSAC'23)*, Tromsø, Norway.

20/06/2023 "**The generating function of DYZ-like numbers is algebraic**",
at Foundations of Computational Mathematics *(FoCM'23)*, Paris, France.

07/03/2023 "**Beating binary powering for computing the Nth power**",
at Journées Nationales de Calcul Formel *(JNCF'23)*, Luminy, France.

13/12/2022 "**Algebraic solutions of continuous and discrete differential equations**",
at *Enumerative Combinatorics*, Oberwolfach, Germany.

28/11/2022 "**Creative Telescoping for the Canham model in genus one**",
at Combinatorics and Arithmetic for Physics *(CAP'22)*, Bures-sur-Yvette, IHES, France.

14/09/2022 "**Hypergeometric diagonals and a step towards Christol's conjecture**",
at *Journées du GDR EFI 2022*, Toulouse, France.

06/07/2023 "**Solving Rupert's Problem Algorithmically**" (Poster presentation),
at Internat. Symposium on Symbolic & Algebraic Computation *(ISSAC'22)*, Lille, France.

29/03/2022 "**Conjecturing & proving that the g.f. of the Yang-Zagier numbers is algebraic**",
at Functional Equations in LIMoges *(FELIM'22)*, Limoges, France.

28/02/2022 "**The Art of Educated Algorithmic Guessing**",
at Journées Nationales de Calcul Formel *(JNCF'22)*, Luminy, France.

01/12/2021 "**How to conjecture & prove that the g.f. of the Yang-Zagier numbers is algebraic**",
at *Combinatorics and Arithmetic for Physics (CAP'21)*, Bures-sur-Yvette, IHES, France.

03/11/2021 "**The Art of Algorithmic Guessing (in gfun)**",
at *Maple Conference*, Virtual, Online.

20/08/2021 "**The Generating Function of the Yang-Zagier Numbers is Algebraic**",
at *Applied Algebraic Geometry (AG'21)*, Virtual, Online.

26/07/2021 "**Diagonals and Hypergeometric Functions**",
at *Applications of Computer Algebra (ACA'21)*, Toulouse, France.

01/03/2021 "**Computing the N-th term of a q-holonomic sequence**",
at Journées Nationales de Calcul Formel *(JNCF'21)*, Luminy, France.

Additional invited talks at seminars & workshops

2026 "**Der Noperteder, die Form die es nicht geben dürfte**",
at *TU4Math – Popular Mathematics Talk*, TU Wien, Vienna, Austria.

2026 "**On Rupert's problem**",
at *Mathematical Colloquium*, University of Vienna, Austria.

2025 "**On Rupert's problem**",
at *Mathematical Colloquium*, University of Mainz, Germany.

09/11/2023 "**Biomembranes and creative telescoping**",
at *Seminar Algebra and Discrete Mathematics*, Linz, Austria.

13/10/2023 "**Rupert or not Rupert?**"
at CADE Kirchberger workshop, Kirchberg, Austria.

21/03/2023 "**Algebraic solutions of DDEs with one catalytic variable**",
at *AG Diskrete Mathematik*, Vienna, Austria.

27/02/2023 "**Algebraic solutions of DDEs**",
at *MathExp Seminar*, Saclay, France.

19/02/2023 "**Creative telescoping for periods**",
at *Workshop on Fuchsian ODEs: "Periods"*, Lisbon, Portugal.

02/12/2022 "**Algebraic functions and integer sequences**",
at *Kirchberger Workshop*, Kirchberg, Austria.

14/05/2022 "**On Yang-Zagier Numbers**",
at *Workshop on Fuchsian ODEs*, Lisbon, Portugal.

03/05/2022 "**The generating function of the Yang-Zagier Numbers is algebraic**"
at *LFANT Seminar*, Bordeaux, France.

20/04/2022 "**Rupert or not Rupert?**"
at *VSM Student Retreat*, Yspertal, Austria.

01/12/2022 "**Algebraicity of solutions of functional equations with one catalytic variable**",
at *Algebraic and Enumerative Combinatorics*, Waterloo, Canada.

22/10/2021 "**Algorithms for proving algebraicity**",
at *Workshop: Fuchs'sche Differentialgleichungen*, Steyr, Austria.

25/09/2020 "**On a class of hypergeometric diagonals**",
at *Transcendance et combinatoire*, Virtual, Online.

10/03/2022 "**The mysterious Apéry numbers**"
at *VSM Student Retreat*, Yspertal, Austria.

17/02/2020 "**Algebraic Power Series**",
at *Workshop on the arithmetic of ODEs*, Lisbon, Portugal.

21/01/2020 "**Some mathematical riddles**",
at *Workshop: Algebraic Geometry in Austria*, Seewalchen am Attersee, Austria.

25/10/2019 "**Künstliche Neuronale Netze**",
at *Pro Scientia*, Vienna, Austria.

21/02/2019 "**Representation of Algebraic Power Series as Diagonals**",
at *Workshop: Algebraic marvels in differential equations*, Lisbon, Portugal.

17/01/2019 "**Some words on the Queuing Theory**",
at *Wesenufer Workshop*, Wesenufer, Austria.

Further participation at conferences and other scientific events

March 2026 Conference: *Journées Nationales de Calcul Formel (JNCF'26)*, Luminy, France.

December 2024 Symposium: *34. SafeTRANS Industrial Day*, Munich, Germany.

September 2023 Summer school: *"Pro Scientia Sommerakademie"*, Puchberg, Austria.

July 2023 Conference: *FPSAC 2023*, Davis, USA.

January 2023 Conference/workshop: *Atelier E-fonctions, G-fonctions et Périodes*, Paris, France.

September 2022 Summer school: *VSM Summer School*, Techendorf am Weißensee, Austria.

April 2022 Conference: *SLC (87)*, Saint-Paul-en-Jarez, France.

March 2022 Conference: *Journées ALEA*, Luminy, France.

September 2021 Summer school: *VSM Summer School*, Techendorf am Weißensee, Austria.

September 2021 Summer school: *"Pro Scientia Sommerakademie"*, Raabs, Austria.

July 2021 Conference: *ISSAC 2021*, Saint Petersburg, Russia (Online).

June 2021 Conference: *Lattice Paths, Combinatorics and Interactions*, Marseille, France.

June 2021 Conference: *EJCIM 2021*, Virtual, Online.

March 2021 Conference: *Journées ALEA*, Luminy, France.

July 2020 Conference: *ISSAC 2020*, Kalamata, Greece (Online).

September 2019 Summer school: *"Pro Scientia Sommerakademie"*, Bozen, Italy.

July 2019 Workshop: *"Lüsens Workshop"*, Lüsens, Austria.

May 2019 Conference: *"Transient Transcendence in Transylvania"*, Brașov, Romania.

September 2018 Summer school: *VDS Summer School*, Techendorf am Weißensee, Austria.

September 2018 Conference: *"Data Science in Finance with R"*, Vienna, Austria.

September 2018 Summer school: *"Pro Scientia Sommerakademie"*, Bad Aussee, Austria.

Peer Review Activities and other scientific activities

2025	Referee for *Experimental Mathematics*
2025	Referee for *The American Mathematical Monthly*
2025	Reviewer for a *WTZ* (Scientific & Technological Cooperation) application
2023	Referee for *The Fibonacci Quarterly*
2023	Referee for *Geometriae Dedicata*
2022 – 2023	Reporter for the Oberwolfach Report 2250 on *Enumerative Combinatorics*
2022	Referee for *Research in Number Theory*
2021 – 2022	Referee for *ISSAC*
2021 – 2022	Referee for *European Journal of Combinatorics*
2021	Referee for *The American Mathematical Monthly*
2021	Referee for *Experimental Mathematics*

Languages

Russian (native), German (native), English (fluent), French (elementary)